

THEORY OF COMPUTATION	
Course Code: AD44	Credits: 2:1:0
Pre – requisites: Basic Mathematics	Contact Hours: 28L+ 14T
Course Coordinator: Dr. Vinay T R	

Course Content

Unit I

Introduction to Finite Automata, structural representations, automata and complexity, the central concepts of automata theory, deterministic finite automata, nondeterministic finite automata, an application of finite automata, finite automata with epsilon transitions.

- Pedagogy/Course delivery tools : Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
- <https://nptel.ac.in/courses/10610408>
- Impartus recording: <https://a.impartus.com/ilc/#/course/81462/295>

Unit II

Regular expressions, finite automata and regular expressions, applications of regular expressions, proving languages not to be regular, closure properties of regular languages: Reversal and Properties under Boolean Operations.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
- <https://nptel.ac.in/courses/10610408>
- Impartus recording: <https://a.impartus.com/ilc/#/course/81462/295>

Unit III

Context—free grammars, parse trees: Constructing and Yield of a Parse tree, applications, ambiguity in grammars and languages, definition of the pushdown automata, the languages of a PDA.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104145>
- <https://nptel.ac.in/courses/10610408>
- Impartus recording: <https://a.impartus.com/ilc/#/course/81462/295>

Unit IV

Deterministic Pushdown Automata, normal forms for CFG: Chomsky Normal Form, the pumping lemma for CFGs, Intersection of CFL and Regular Language.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
- <https://nptel.ac.in/courses/10610408>
- Impartus recording: <https://a.impartus.com/ilc/#/course/81462/295>

Unit V

The Turing machine, Extensions to the basic Turing machine, Turing machine and computers, Undecidability: An Undecidable Problem that is RE ,Definition of Post's Correspondence problem.

- Pedagogy/Course delivery tools: Chalk and talk, Power Point Presentation, Videos
- Links: <https://nptel.ac.in/courses/106104148>
- <https://nptel.ac.in/courses/10610408>
- Impartus recording: <https://a.impartus.com/ilc/#/course/81462/295>

Suggested Learning Resources

Text Book:

1. John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman: Introduction to Automata Theory, Languages and Computation, 3rd Edition, Pearson Education, 2011.

Reference Books:

1. John C Martin: Introduction to Languages and Automata Theory, 3rd Edition, Tata McGraw-Hill, 2007.
2. Michael Sipser: Introduction to the Theory of Computation, 3rd Edition,
3. Thompson Course Technology, Boston, MA and Cengage Learning India Pvt. Ltd., 2014.

Course Outcomes (COs):

At the end of the course, the student will be able to:

1. Explain the basic concepts of formal languages and finite automata. (PO-1, PSO-1)
2. Construct automata to accept strings from a specified language. (PO-1,2,3, PSO-1)
3. Convert among equivalently powerful notations for a language, including among DFAs, NFAs, and regular expressions, between PDAs, CFGs and normal forms of CFGs. (PO-1,2,3, PSO-1)
4. Prove the various closure and decision properties of formal languages. (PO-1,2, PSO-1)
5. Explain the concepts of Undecidability, RE languages and Post Correspondence problem. (PO-1,2, PSO-1,3)

Course Assessment and Evaluation:

Continuous Internal Evaluation (CIE): 50 Marks		
Assessment Tools	Marks	Course Outcomes (COs) addressed
Internal Test-I (CIE-I)	30	CO1, CO2, CO3
Internal Test-II (CIE-II)	30	CO4, CO5
Average of the two CIE shall be taken for 30 marks		
Other Components		
Quiz -I	10	CO1, CO2, CO3
Assignment	10	CO3, CO4, CO5
The Final CIE out of 50 Marks = Average of two CIE tests for 30 Marks+ Marks scored in Quiz-I +Marks scored in Assignment		
Semester End Examination (SEE)		
Course End Examination (Answer One full question from each Unit- Internal Choice)	100	CO1, CO2, CO3, CO4, CO5